Framework-Decoupled MoE Load Balancing for High-Performance LLM Inference

Problem statement:

MoE inference performance is limited by expert-parallel load imbalance. Static expert placement helps, but live batches can still overload a subset of EP ranks, shared-expert work can remain poorly distributed, and replicated routed experts need per-batch routing decisions. At the same time, each new load-balancing idea is usually integrated separately into SGLang, vLLM, TensorRT-LLM, or other runtimes, creating duplicated engineering work and slowing product adoption.

Importance:

Large MoE models are becoming central to high-throughput LLM serving. A portable MoE load balancer can improve NVIDIA GPU utilization, reduce serving cost, and make new MoE algorithms easier to productize across inference frameworks. The work also creates a common path from research prototypes to production stacks such as SGLang, vLLM, and TensorRT-LLM, reducing the N algorithms x M frameworks integration cost.

Prior work:

The design builds on DeepSeek EPLB/LPLB, SGLang DeepEP and Waterfill work, vLLM EPLB support, TensorRT-LLM Wide-EP/online EPLB, and recent research such as topology-aware EPLB, CRAFT, MoETuner, Metro, Libra, and PROBE. These systems demonstrate strong individual techniques, but their implementations are tied to specific runtime APIs. This submission focuses on turning those ideas into a framework-neutral load-balancing layer with thin runtime adapters.

Technical solution:

We define a framework-decoupled MoE load balancer with three orthogonal layers. L1 performs static expert placement and replica layout planning. L2 performs per-batch token routing while preserving the router top-k and only changing physical execution choices. L3 reserves a future surface for prediction-driven expert prefetch or runtime migration. The core API uses framework-neutral contracts such as PlacementProvider.snapshot, PlacementPlanner.plan, and TokenRouter.route_tokens, while adapters translate to SGLang, vLLM, or TensorRT-LLM-specific data structures. The SGLang POC integrates L1 EPLB, Waterfill, and LPLB, and validates the adapter boundary end to end.

Quantitative benefits:

On two-node H20 DeepSeek-V3/R1-style serving workloads, the latest SGLang/DeepEP measurements show Waterfill improving total throughput by +1.48% to +4.66% across MMLU, GPQA, and GSM8K. LPLB improves redundant-EPLB configurations by up to +7.34%, with the strongest result on GSM8K red32. On DeepSeek V4 Flash, Waterfill improves the best measured point from 49,253 tok/s to 51,677 tok/s (+4.92%). These gains preserve model semantics because the router top-k is unchanged; only the physical shared-expert rank or valid routed-expert replica is selected differently.